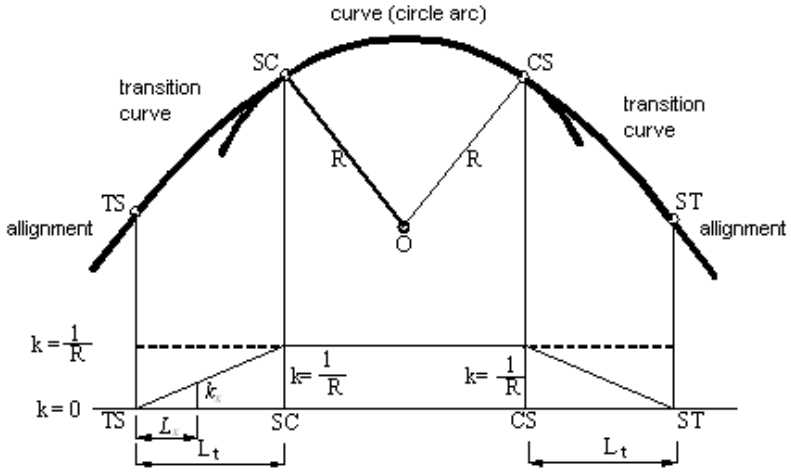
7.7 TRANSITION CURVES

Additional highway components that are placed between the alignment and the horizontal curve in order to resolve the sudden effect of the centrifugal force are named **transition curves**. The effect of the centrifugal force, which emerges suddenly because of the different curvatures of the alignment and the horizontal curve, is countered by these components by spreading through a (L_t) length. The alignment and the horizontal curve circle arc are conjoined by the transition curve. Thus, the transition curve has to provide the following conditions.

1. The start of the transition curve is tangent to the alignment, while its end is tangent to the horizontal curve.
2. The radius of curvature ($R_t=\infty$) is infinite at the beginning of the transition curve, while it is equal to the radius of the horizontal curve ($R_t=R$) at the end. This decrease in the radius of curvature, or in other words, the increase in the curvature (k), has to be linear through the curve, as seen in the figure.
3. The vehicle must be able to continue its course with the constant velocity, V, through the transition curve.
4. When the steering wheel of the vehicle is rotated with a constant angular velocity of ω, the highest rotation angle has to be obtained at the end of the transition curve or, in other words, at the beginning of the horizontal curve.

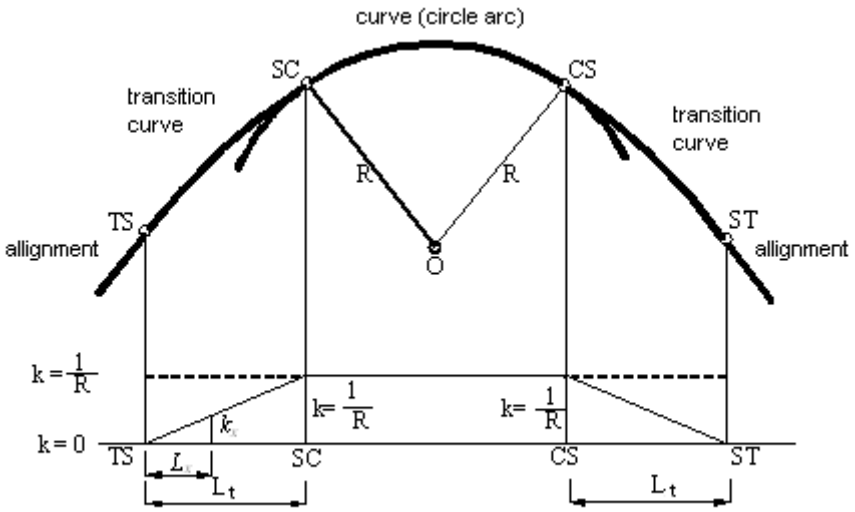


The curvature of any point L_x which is located after the TS start point of the transition curve, can be calculated as:

$$\frac{k_x}{L_x}=\frac{k}{L_t} \qquad \rightarrow \qquad k_x=\frac{k}{L_t}\cdot L_x$$

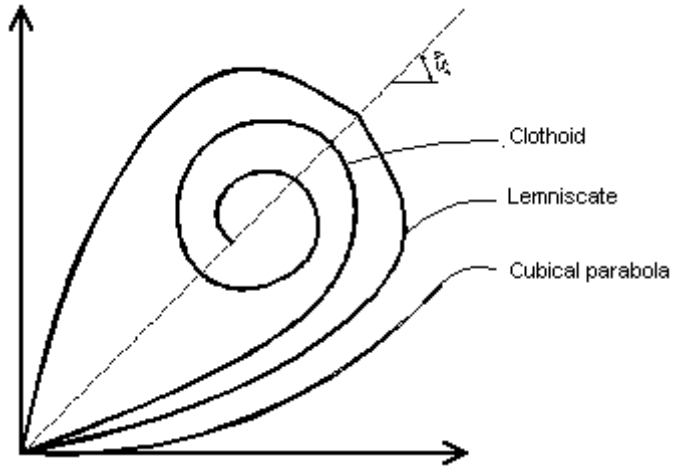
At the end of the curve, or at the point SC, the curvature reaches the value of k=1/R, so k_x can be defined as:

$$k_x = \frac{L_x}{L_t \cdot R}$$



•

As the (L_t.R) term is a constant on the statement, it shows that the curvature increases linearly. Various types of transition curves that fit this description are clothoid, lemniscate, and cubical parabola. Also, the circle arc with a radius of 2R can be used as a transition curve.



7.7.1 Transition curve length

When determining the transition curve length, the jerk has to be checked according to the dynamics of vehicle movements.

The latitudinal acceleration value becomes zero at the end of the alignment, or in other words, at the beginning of the transition curve, while it becomes (v^2/R) at the end of the transition curve or, in other words, at the beginning of the horizontal curve.

The change of the latitudinal occurs during $t=L_t/v$.

Accordingly, if the jerk formula is written as:

$$p' = \frac{\frac{v^2}{R}}{\frac{L_t}{v}} = \frac{v^3}{R \quad L_t}$$

is obtained. If the velocity is written in terms of km/h (v=V/3,6) as the project velocity, the transition curve length can be calculated as:

$$L_t = \frac{V_p^3}{46{,}7 \cdot R \cdot p'}$$

The transition curve length can be calculated for a horizontal curve with a specific project velocity and a radius.

Transition curve length for p' = 0,40 (m/sec^3), $L_t = \frac{V_p^3}{18{,}7 \cdot R}$

Transition curve length for p' = 0,50 (m/sec^3), $L_t = \frac{V_p^3}{23{,}4 \cdot R}$

Transition curve length for p' = 0,60 (m/sec^3), $L_t = \frac{V_p^3}{28{,}0 \cdot R}$

7.7.2 The geometrical properties of the clothoid

The geometrical properties of the transition curve have to be known comprehensively in order to apply them properly on the land. Only the properties of the clothoid will be mentioned in this lecture.

Clothoid is the most preferred transition curve in highway design. If the radius of curvature on any point x of the clothoid is R_x, and the distance of this point to the start of the curve is L_x, the relation between them can be written as:

$$R_X \quad L_X = A^2$$

A is named as the flatness of the spiral and its unit is in meters. On the intersection point of the clothoid with the horizontal curve, $R_x=R$ and $L_x=L_t$. In this case

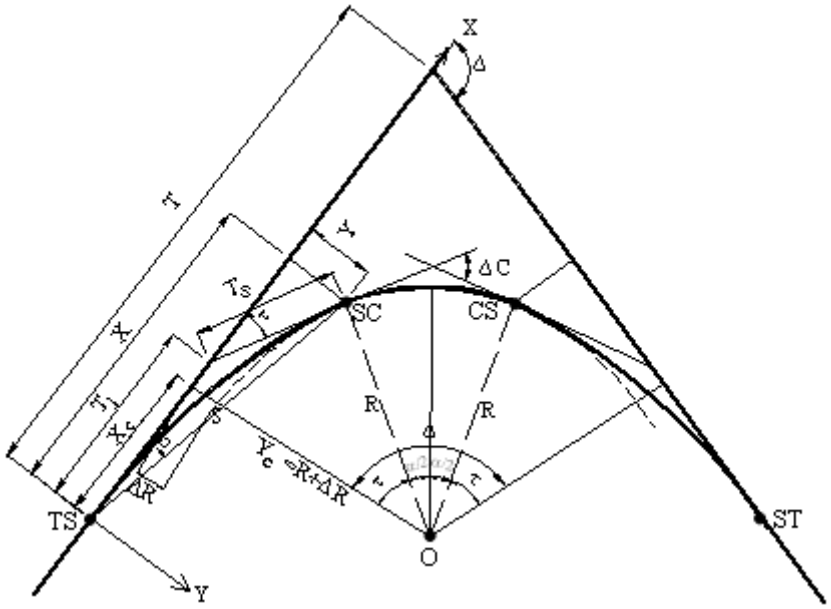
$$R \cdot L_t = A^2$$

A more generalized formula can be written. In this case, the flatness of a clothoid doesn't change or in other words, it is constant. If the flatness of the clothoid is changed, its length also changes.

There's a single shape of the clothoid; however, it changes with respect to its members. In other words, the length of the clothoid can be changed according to the value of the A parameter. In this regard, the flatness of the clothoid can also be defined as the enlargement factor of the clothoid.

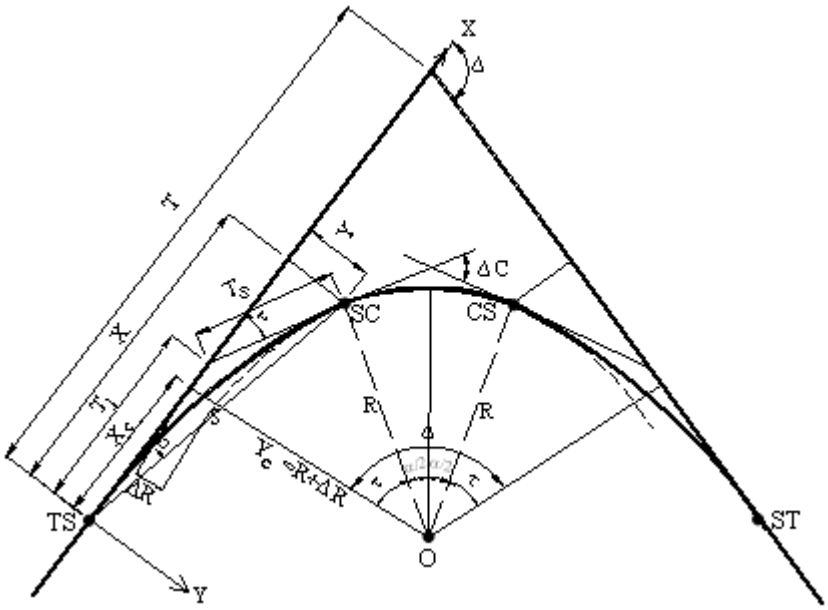
The clothoid with the flatness value of A = 1 is named as the unit clothoid. Unit clothoid is multiplied with various A parameters to obtain many various-size clothoids. Tables have been prepared to use on horizontal curve design by using this property of the clothoid.

The elements of the clothoid applied symmetrically to a horizontal curve can be seen in the figure below. Distances l, r, Δr, x_c, x, y, t_s, t_l can be obtained directly from the table for the clothoid with the flatness of A, and the other elements can be calculated as: L=A.l, R=A.r, ΔR=A.Δr, X_m=A.x_c, X=A.x, Y=A.y, T_s=A.t_s, T_l=A.t_l. In this case, as the ratio between the elements does not change, the values of the angles also remain the same.



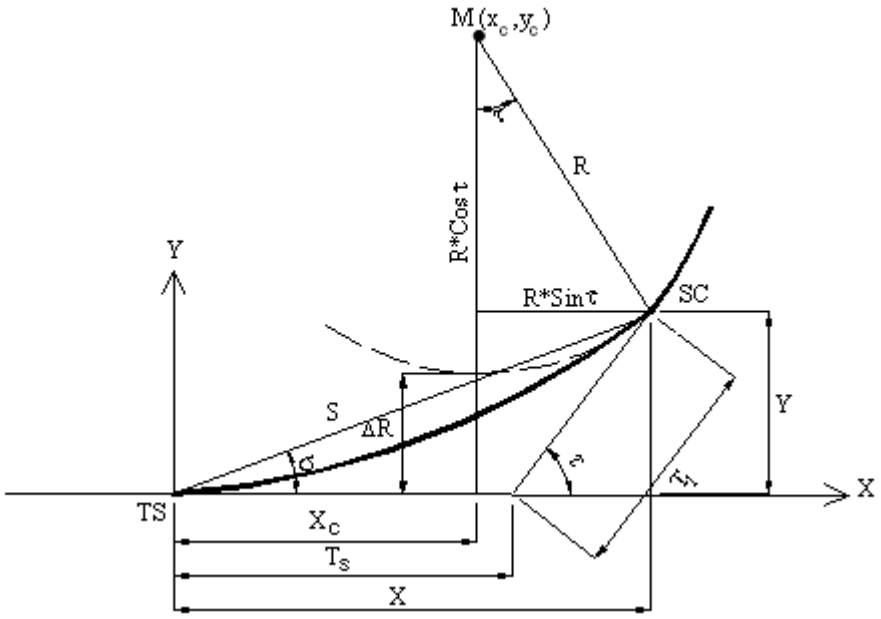
The clothoid is a spiral having a specific mathematical function. The whole spiral is not used in the highway design. Only the part of it that is suitable for a transition curve according to the given properties is being used.

When the elements of the clothoid are calculated, the tangent of the horizontal curve is accepted as the X (abscissa), while the perpendicular axis is accepted as the Y (ordinate). The calculation and the drawing of the elements are made with respect to this cartesian coordinate system. Integrals are used in order to calculate the cartesian coordinates of the clothoid. The first three terms of an expansion of a series are taken into account. Clothoid tables are developed in order to avoid remaking these calculations each time.



The elements of the clothoid can be listed as:

- L : Length of the clothoid (also can be notation as L_t as the transition curve length) (m),
- X : The abscissa of the endpoint of the clothoid (SC) with respect to its starting point (TS) (m),
- Y : The ordinate of the endpoint of the clothoid (SC) with respect to its starting point (TS) (m),
- τ : The tangent angle of the clothoid (radians),
- X_c : The abscissa of the horizontal curve center (m),
- Y_c : The ordinate of the horizontal curve center (m),
- T_l : The length of the longer tangent of the clothoid (m),
- T_s : The length of the shorter tangent of the clothoid or the distance between the tangent of the curve to the common tangent point SC (m),
- ΔR : Displacement of the horizontal curve (m),
- S : Clothoid chord (m),
- σ : Direction angle of the clothoid chord (degree),
- : The tangent length of the clothoid or the distance between the start and the PI of the horizontal curve (m),
- ΔC : Divergence angle of the shifted curve (degree),
- Lc : Length of the shifted curve (m) ($Lc=CS_{km}-SC_{km}$).



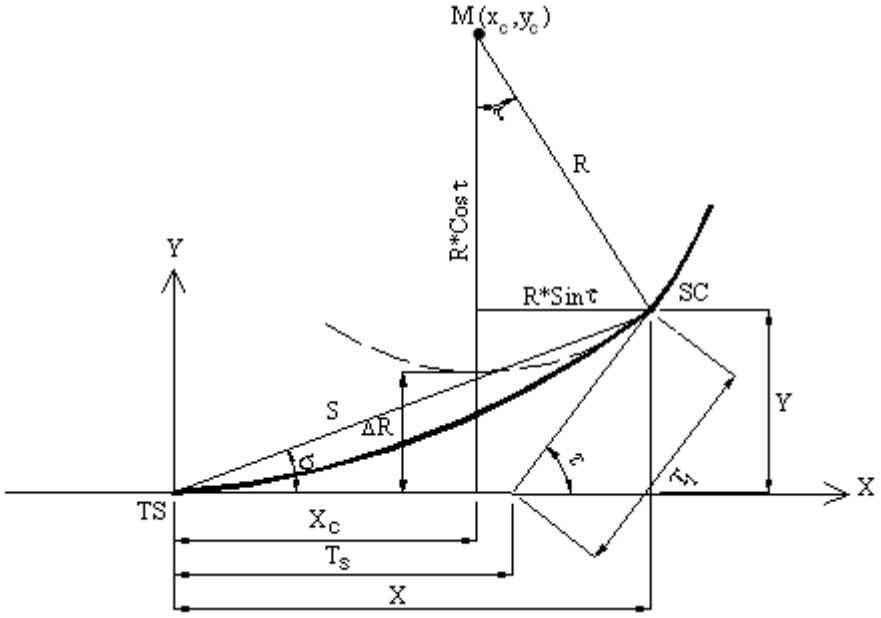
The critical cross-sections of the clothoid have a special notation on the plan. They can be listed as:

TS : The starting point of the clothoid on the entrance of the horizontal curve (**T**angent-**S**piral)

SC : The endpoint of the clothoid on the entrance of the horizontal curve (**S**piral-**C**urve)

CS : The starting point of the clothoid on the exit of the horizontal curve (**C**urve-**S**piral)

ST : The endpoint of the clothoid on the exit of the horizontal curve (**S**piral-**T**angent)



The following formulas are used in calculating the elements of the clothoid.

$$X = L \quad \frac{L^5}{40 \quad A^4} + \frac{L^9}{3456 \quad A^8} \quad (m)$$

$$Y = \frac{L^3}{6 \quad A^2} \quad \frac{L^7}{336 \quad A^6} + \frac{L^{11}}{42240 \quad A^{10}} \quad (m)$$

$$\tau = \frac{L}{2 \cdot R} \quad (\text{radians})$$

$$X_c = X - R \cdot \sin\tau \quad (m)$$

$$Y_c = Y + R \cdot \cos\tau \quad (m)$$

$$T_l = X - Y \cdot \text{cotg}\tau \quad (m)$$

$$T_s = \frac{Y}{\sin\tau} \quad (m)$$

$$\Delta R = Y_c - R \quad (m)$$

$$S = \sqrt{X^2 + Y^2} \quad (m)$$

$$\sigma = \text{arctg}\left(\frac{Y}{X}\right) \quad (\text{degree})$$

The divergence angle of the horizontal curve changes as the clothoid is placed, so the length of the circle arc also changes. The new values can be calculated as:

$$\overline{TS} = X_c + (R + \Delta R) \cdot \text{tg}\frac{\Delta}{2} \quad (m)$$

$$\Delta C = \Delta - 2 \cdot \tau \quad (\text{degree})$$

$$Lc = \frac{2 \cdot \pi \cdot R \cdot \Delta C}{360} \quad (m)$$

7.7.3 Selection of the Clothoid Flatness

The clothoid flatness can be selected in two ways. First by using the clothoid template and the other by calculations.

The usage of the clothoid template is known as the graphical method. In this method, templates having various A flatness values and radius of curvature are used. The template is placed on the route axis to select the most appropriate clothoid. The radius that fits the endpoint of the clothoid is selected as the curve radius. This method is no longer used as the computer technology is widespread.

In the calculation method, the maximum jerk value is taken into account in terms of comfort. The A flatness value is selected with respect to the following criteria and the necessary clothoid length is calculated afterwards:

- Optical condition:

$$\frac{R}{3} \leq A \leq R$$

- Dynamical condition:

$$A_{min} = 0{,}19\sqrt{V_p^3}$$

- Superelevation condition:

$$A_{min} = \sqrt{\frac{b}{2} \cdot \frac{(q_{min} + q_{max})}{0{,}005} \cdot R}$$

Clothoid flatness values calculated by using the dynamical condition and the superelevation condition are between the value obtained by the optical condition and the limit value. On the other hand, the clothoid flatness value obtained from the superelevation condition is usually higher than the value obtained from the dynamical condition. This condition also affects the clothoid lengths for the same radius. In other words, the length of the transition curve calculated according to the clothoid flatness selected using the superelevation condition is longer than the one selected using the dynamical condition. On low-standard highways, due to the topographical constraints, enough distance to apply the calculated transition curve length according to the superelevation condition may not be availible. For this reason, on low-standard highways, clothoid flatness, A is selected according to the first two conditions, while on higher-standard roads, the third condition is also taken into account. Finally, the clothoid length is calculated by using the following formula after the flatness is determined:

$$L_t = \frac{A^2}{R}$$

The clothoid length has to be compatible with the superelevation attainment length. For this reason,

$$L_t \geq L_{se,min} = 45m$$

condition has to be fulfilled. To be able to apply a clothoid of this length, the horizontal curve circle arc also should have a minimum length that can be checked as:

$$\frac{L_t}{2} \leq \frac{L}{2}$$

Finally the following comfort criterion, jerk is checked to finalize controls:

$$p' = \frac{V_p^3}{46{,}7 \cdot R \cdot L_t} \leq 0{,}6m/\sec^3$$

7.8 WIDENING OF THE HORIZONTAL CURVES

The orbits of the frontal and the rear tires are different while the vehicles are traveling on a horizontal curve. Frontal tires turn by drawing a wider arc than the rear tires. When drivers enter a horizontal curve, they have a tendency to recede from the inside edge and to get closer to the road axis. Especially on undivided two-lane highways, this condition results in getting closer to the opposite lane or even violating it. To resolve this situation, which endangers driving safety on low-radius horizontal curves, the platform is widened. To calculate the amount of the widening, the figure is used. As the rotation of the vehicle on the horizontal curve is a circular motion, by using the ABO right triangle,

$$R^2 = R^2 - 2Rb_w + b_w^2 + l^2$$
$$R^2 = (R - b_w)^2 + l^2$$
$$2Rb_w - b_w^2 = l^2$$
$$b_w(2R - b_w) = l^2$$

As the b_w term on the left side of the equation is too small according to the 2R term, it can be neglected.

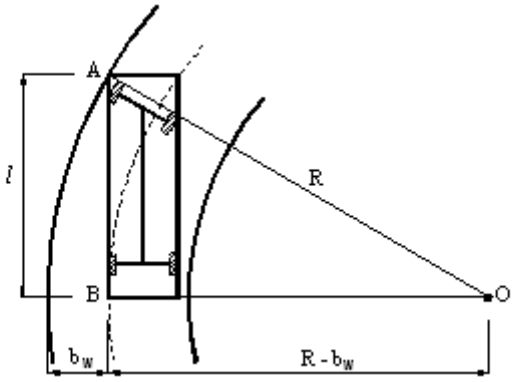
In this case, the widening amount can be calculated as:

$$b_w = \frac{l^2}{2 \cdot R}$$

b_w : The widening amount of a single lane (m),

R : The radius of the orbit of the outer edge of the vehicle (m). Horizontal curve radius is used in calculations.

l : Vehicle length(m)



The widening amount calculated is appropriate for highways with a low project velocity.

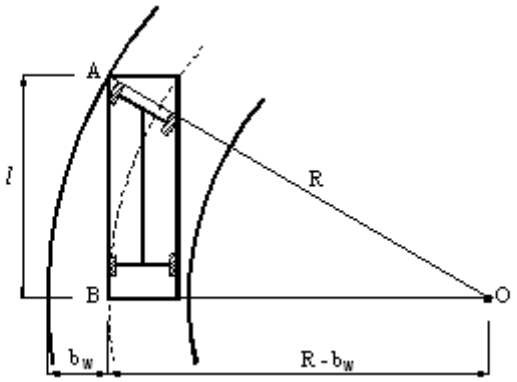
As the velocity increases, a small orbital change on the vehicle tyre results in a high lateral displacement. In this case, the widening has to be increased in order to prevent the vehicle from violating the opposite lane. So, the formula has to be corrected with an empirical velocity term to take the form of:

$$b_w = \frac{l^2}{2 \cdot R} + \frac{0{,}05 \cdot V_p}{\sqrt{R}}$$

The value to be used for the *l* distance is determined according to the dominant traffic conditions. If the heavy vehicle ratio is high, *l*=10~12 m can be accepted. The total widening amount for a highway having *n* lanes is calculated as:

$$b_W = \frac{n}{2} \frac{l^2}{R} + \frac{0{,}05 \quad V_p}{\sqrt{R}}$$

The widening can be done in two ways: one equally on both sides of the road axis and the other only on the inner side. In the first method, half of the calculated bw amount is given on the inner side and the other half to the outer side. In the second method, the whole b_w amount calculated is given on the inner side. However, the axis line should still be drawn as centering the platform.



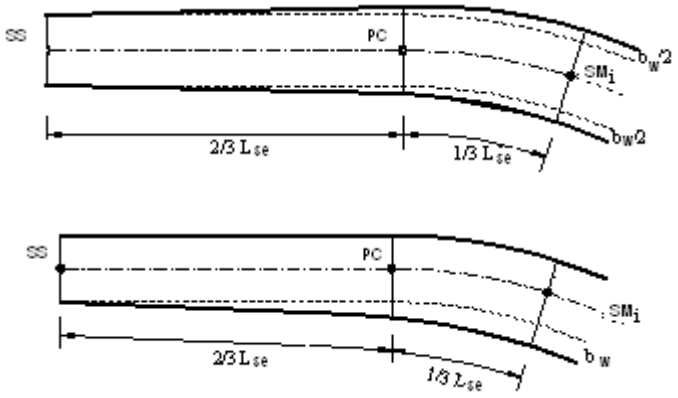
7.8.1 Widening the curves without a transition curve

The widening is applied through the superelevation length on curves without a transition curve. Its change is linear just like the superelevation. The widening amount, which is zero at the start of the superelevation (SS), reaches its maximum value where the superelevation is also maximum (SM_i).

The same amount continues through the circle arc. On the exit of the horizontal curve, the widening decreases through the superelevation length to reach the value of zero on point SE.

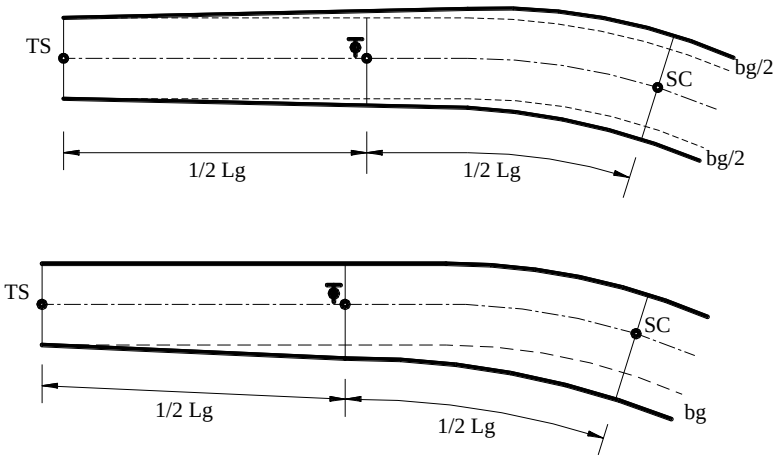
The maximum curve radius for which a widening is needed is R< 200 m. For radii higher than 200 m, the widening can be neglected and is not applied. The upper limit of the widening is $b_{w,max}$=2,0 m. When the superelevation distances are being calculated for a widened horizontal curve, the widening amount has to be added to the platform width. As an example, the widening amount for the SM_i point that the superelevation reaches the maximum amount can be calculated as:

$$h_{max} = q_{max} \cdot \frac{(b + b_w)}{2}$$



7.8.2 Widening the curves with a transition curve

The widening is applied through the transition curve length on the curves with a transition curve. The effects of centrifugal force create maximum discomfort in this section. The radius and the widening amount constraints are also valid in this case. The widening amount, which is zero at the start of the transition curve (TS), increases linearly to reach the maximum amount on the point SC. The widening starts to decrease symmetrically on the point CS and becomes zero on the ST.



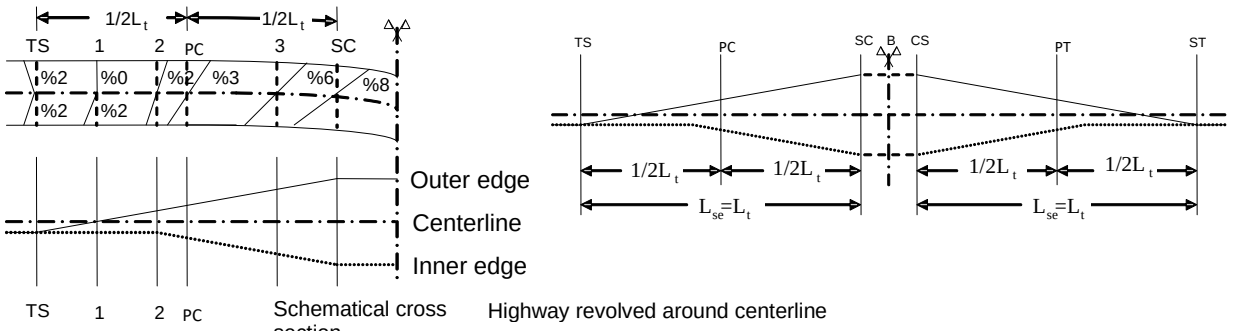
7.9 Implementation of the superelevation on horizontal curves with a transition curve

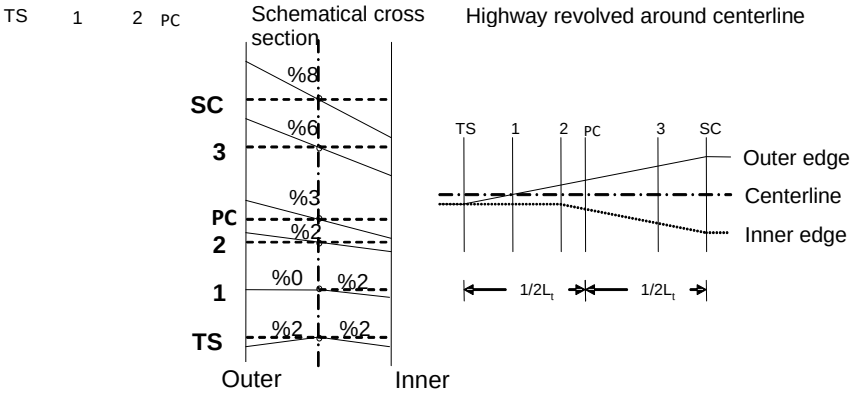
On horizontal curves with a transition curve, the increase of the superelevation from the crown slope to the maximum amount is done through the transition curve length. In other words, superelevation change is applied through the transition curve. Superelevation application methods are also the same in this situation as **highways revolve around the centerline, inner edge,** or **outer edge**.

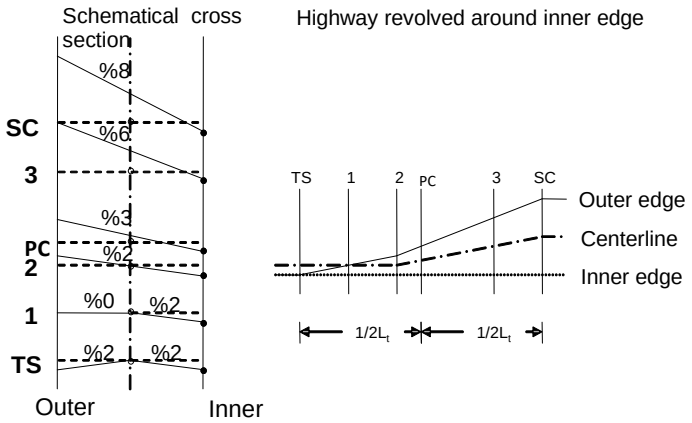
Additionally, according to the importance and the type of the highway, the application can be done by using a ramp or not.

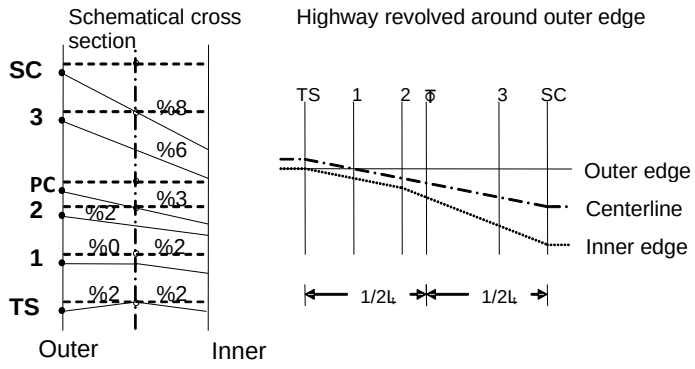
7.9.1 Superelevation application without using a ramp

In this application method, the crown slope starts to change 1/2Lt before the PC and reaches the maximum amount of 1/2Lt after the PC by linear increase. In other words, the slope starts to change on TS and reaches the maximum amount on CS. Symmetrically, the slope starts to decrease on CS to reach the roof slope on ST to finalize the superelevation application. Application of superelevation without using a ramp, taking L_t=45 m, is shown in the figures.









7.9.2 Superelevation application using a ramp

As the curvature starts to increase as entering the transition curve, the centrifugal force also starts to affect the vehicle. To counter these negative effects, the highway is desired to reach the single slope on this point. The change of the roof slope to reach the single slope on the point TS has to be done within a distance. This distance is named as the **ramp length**. On the other hand, by the application of the ramp length, the effect of the inverse superelevation affecting the vehicles moving in the opposite direction on the outer lane is prevented for safe travel.

The superelevation, which reaches the single slope at the start of the transition curve, increases linearly to reach the maximum value afterward. On the calculation of the ramp length, different conditions arise with respect to the superelevation applied as the highway revolves around the centerline, inner edge, or outer edge.

The superelevation diagram drawn by taking ramp length into account differs from those in which ramp length is not considered. The change on the edge, which is elevated, has to be linear until the end of the transition curve. The superelevation application uses a ramp, and calculations are given in the following slides.

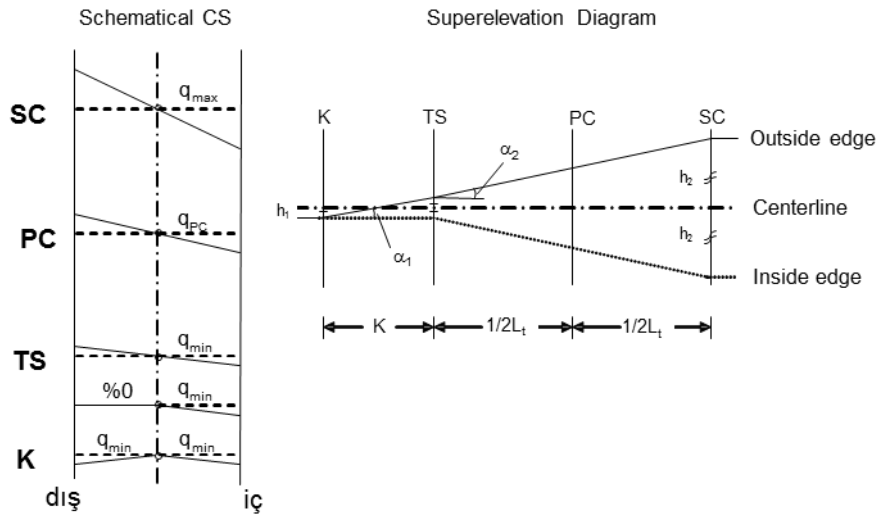
7.9.2.1 The highway revolved around the centerline using a ramp

As seen in the figure, the change in elevation of the outer edge, which starts at point K, continues to increase until point TS. Accordingly, as seen on the superelevation diagram,

$$tg\alpha_1 = tg\alpha_2$$

Using the right triangles, angles can be written with respect to the superelevation height, ramp length, and transition curve length, as seen below. Here h_1 and h_2 are the elevation differences between the constant centerline and the inner/outer edge.

$$tg\alpha_1 = \frac{2 \quad h_1}{K} \qquad tg\alpha_2 = \frac{(h_2 \quad h_1)}{L_g}$$



So, the formula that gives the K ramp length can be obtained as:

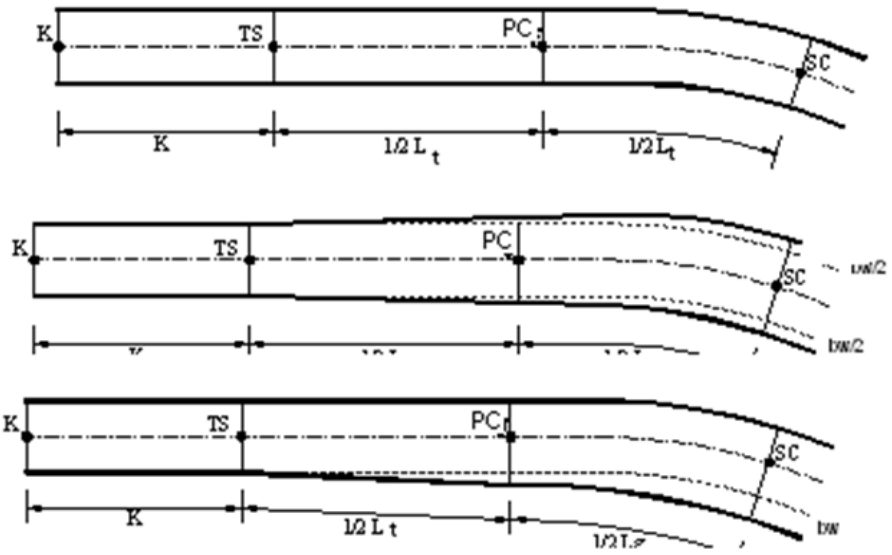
$$\frac{2 \cdot h_1}{K} = \frac{(h_2 - h_1)}{L_t} \qquad K = \frac{2 \cdot h_1 \cdot L_t}{(h_2 - h_1)}$$

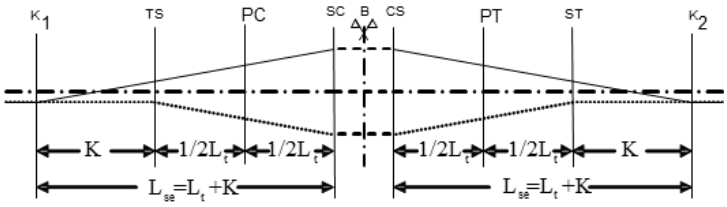
As b is the platform width on the superelevation application as the highway revolved around the centerline,

$$h_{min} = h_1 = q_{min} \frac{b}{2} \qquad h_{max} = h_2 = q_{max} \cdot \frac{b}{2}$$

should not be forgotten. The application of the ramp length, with and without an application of widening, is shown on a schematical plan. If widening is applied, b_w has to be added to the platform width and is calculated.

$$h_{max} = h_2 = q_{max} \cdot \frac{(b + b_w)}{2}$$





7.9.2.2 The highway revolved about the inner edge using a ramp

As seen in the figure, the change in elevation of the outer edge, which starts at point K, continues to increase until point TS. Accordingly, as seen on the superelevation diagram again,

$$tg\alpha_1 = tg\alpha_2$$

Using the right triangles, angles can be written with respect to the superelevation height, ramp length, and transition curve length, as seen below. Here h_1 and h_2 are the elevation differences between the constant inside edge and the outside edge.

$$tg\alpha_1 = \frac{h_1}{K} \qquad tg\alpha_2 = \frac{(h_2 - h_1)}{L_t}$$

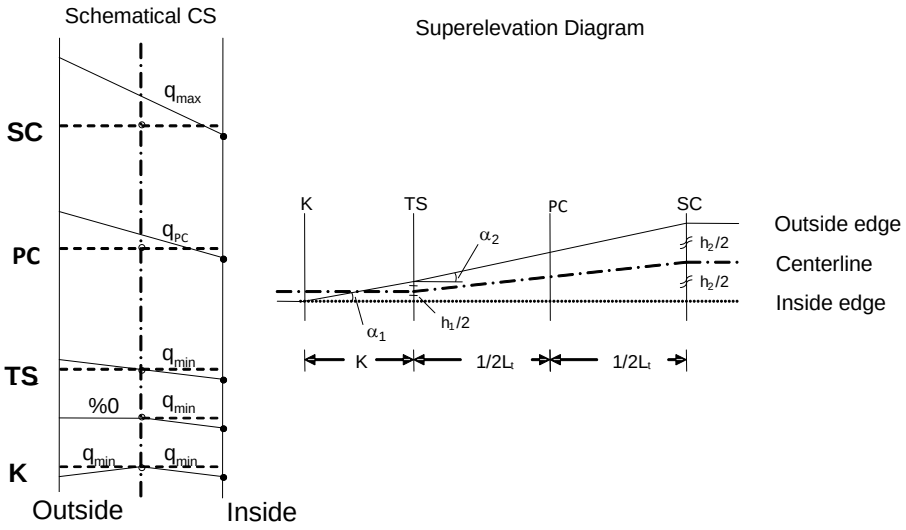
is obtained. As the angles have to be equal,

$$\frac{h_1}{K} = \frac{(h_2 - h_1)}{L_t}$$

can be written. If the K ramp length is taken,

$$K = \frac{h_1 \cdot L_t}{(h_2 - h_1)}$$

is obtained



As b is the platform width on the superelevation application as the highway revolved about the inside edge, superelevation heights can be calculated as:

$$h_{min} = h_1 = q_{min} \quad b \qquad\qquad h_{max} = h_2 = q_{max} \cdot b$$

If the widening is applied, b_w has to be added to the platform width.

7.9.2.3 The highway revolved about the outer edge using a ramp

As seen in the figure, the change in elevation of the inner edge, which starts at point K, continues to decrease until point TS. Accordingly, as seen on the superelevation diagram again,

$$tg\alpha_1 = tg\alpha_2$$

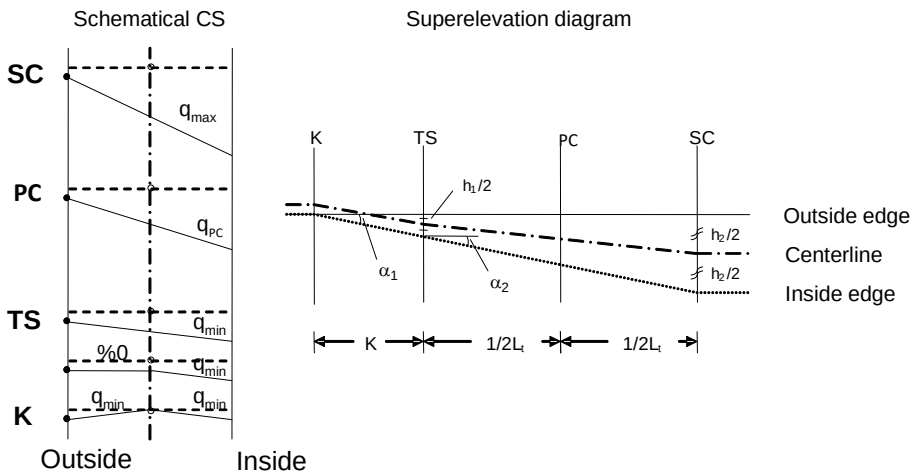
it has to be fulfilled. Using the right triangles, angles can be written with respect to the superelevation height, ramp length, and transition curve length, as seen below. Here h_1 and h_2 are the elevation differences between the constant outside edge and the inside edge.

$$tg\alpha_1 = \frac{h_1}{K} \qquad\qquad tg\alpha_2 = \frac{(h_2 - h_1)}{L_t}$$

is obtained. If these two formulas are equalized, $\frac{h_1}{K} = \frac{(h_2 - h_1)}{L_t}$

can be written. If the K is taken, the ramp length can be calculated as:

$$K = \frac{h_1 \cdot L_t}{(h_2 - h_1)}$$



As b is the platform width on the superelevation application as the highway revolved about the outside edge, superelevation heights can be calculated as:

$$h_{min} = h_1 = q_{min} \cdot b \qquad h_{max} = h_2 = q_{max} \quad b$$

If the widening is applied, b_w has to be added to the platform width.

Conditions encountered on superelevation application

Conditions encountered on superelevation application (Detailed)	Without transition curve			With transition curve		
	Revolved about centerline	Revolved about Inside edge	Revolved about outside edge	Revolved about centerline	Revolved about Inside edge	Revolved about outside edge
No ramp-no widening	x	x	x	x	x	x
No ramp-one sided widening	x	x	x	x	x	x
No ramp-two sided widening	x	x	x	x	x	x
Ramp-no widening				x	x	x
Ramp-one sided widening				x	x	x
Ramp-two sided widening				x	x	x

7.10. SIGHT DISTANCE AT HORIZONTAL CURVES

On the horizontal curves, calculation, and implementation of curve radius, superelevation, and widening of the platform according to parameters like project velocity is obligatory but still not enough to provide a safe and comfortable ride. At curves, sufficient sight distance should be provided for safety.

At curves, minimum sight distances for stopping or overtaking, which were analyzed previously, may not be possible because of obstacles like structures, plantings, cut slopes, etc. In the case that any obstacle prevents the sight distance but makes it impossible to move, the radius of the curve should be increased, and the route should be reassigned. In the case that the curve falls into a cut section, in other words, where the sight distance is prevented by a cut slope, the cut section should be excavated, and the cut slope should be decreased.

The distance that should be kept between the obstacle and the road axes in order to provide the projected sight distance could be calculated using the figure below. Two situations may arise in this process: 1) the curve length is greater than the sight distance, and 2) the curve length is less than the sight distance.

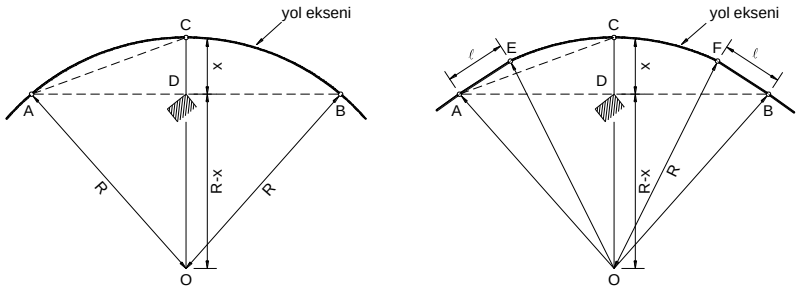
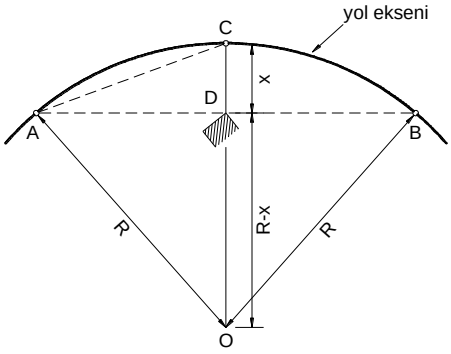


Figure Latitudinal distance at horizontal curves *(Yayla, 2002)*

Situation 1: S<d (curve length is greater than the sight distance)

In the figure below, ACB is assumed to be the required sight distance, and x is the distance between the obstacle and the road axes in a curve with radius R. For simplification, it is assumed that the vehicle rides through the lines AC and CB instead of ACB arc.



Şekil 7.34 Latitudinal distance (x) in a case where S < d *(Yayla, 2002)*

$$\overline{AC}^2 = \overline{AD}^2 + x^2 \qquad (7.91)$$

$$\overline{AD}^2 = R^2 - (R - x)^2 \qquad (7.92)$$

$$\overline{AC}^2 = 2Rx \qquad (7.93)$$

If AC ≅ 1/2. S is taken;

$$\frac{S^2}{4} = 2Rx$$

(7.94a)

$$x = \frac{S^2}{8R}$$

(7.94b)

Situation 2: S>d (curve length is less than the sight distance)

From the figure below, it is assumed that EF = Lc (Curve length),

$$S = d + 2l$$

(7.95a)

$$l = \frac{S - d}{2}$$

(7.95b)

Can be written. Here, l is the length that is distributed evenly to both sides of the curve.

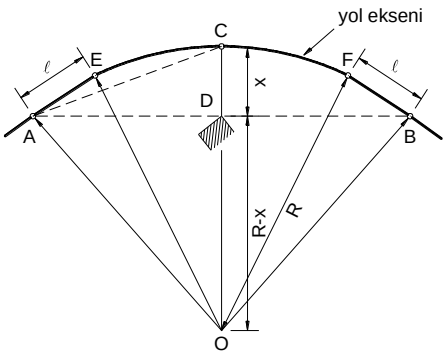


Figure Latidunal distance in case where S > d *(Yayla, 2002)*

On the other hand, from the triangles ACD, ADO, and AEO,

$$\overline{AC}^2 = \overline{AD}^2 + x^2$$

$$\overline{AD}^2 = \overline{AO}^2 - (R - x)^2$$

(7.96)

$$\overline{AO}^2 = l^2 + R^2$$

(7.97a)

$$\overline{AO}^2 = \left(\frac{S - d}{2}\right)^2 + R^2$$

(7.97b)

$$\overline{AD}^2 = \left(\frac{S-d}{2}\right)^2 + R^2 - (R-x)^2$$

(7.98)

$$\overline{AC}^2 = \left(\frac{S-d}{2}\right)^2 + R^2 - (R-x)^2 + x^2$$

(7.99a)

$$\overline{AC}^2 = \left(\frac{S-d}{2}\right)^2 + 2Rx$$

(7.99b)

By taking AC ≅ S/2,

$$\frac{S^2}{4} = \frac{S^2}{4} + \frac{d^2}{4} - \frac{S.d}{2} + 2R.x$$

(7.100a)

$$x = \frac{d(2S-d)}{8R}$$

(7.100b)

can be written.

If d = S is taken, the equation for the first situation will be determined. The stopping sight distance or overtaking distance should be taken as S. If the road is designed as an undivided road, the safety overtaking sight distance should be taken into account.

The figure below shows the plan and profile of a road on which an obstacle exists. As stated before, if the obstacle is a cut section, the slope of the cut could be smoothened, or the curve radius could be increased. As a last solution, the route axis should be shifted.

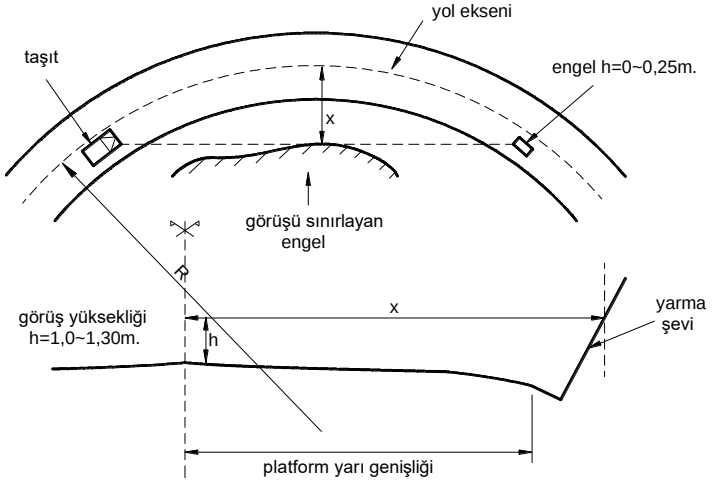


Figure Restricted sight distance at horizontal curves *(Yayla, 2002)*